

27

B

Electrons emitted per unit time:

$$dN_e / dt = I / e = 2.0 \times 10^{-6} / (1.6 \times 10^{-19})$$

$$= 1.25 \times 10^{13}$$

Photons emitted per unit time:

$$dN_p / dt = P / E = 3.1 \times 10^{-3} / (3.11 \times 1.6 \times 10^{-19})$$

$$= 6.23 \times 10^{15}$$

$$\frac{\text{electrons emitted per unit time}}{\text{photons incident per unit time}} = \frac{1.25 \times 10^{13}}{6.23 \times 10^{15}} = 2.0 \times 10^{-3}$$

28

A